

Advection Operator in Incompressible Navier–Stokes (Flux Form)

(Dated:)

I. 1. OVERVIEW

This document explains the implementation of the advection operator using the **flux (divergence) form**:

$$-\nabla \cdot (\mathbf{u}\mathbf{u}) \quad (1)$$

instead of the convective form:

$$(\mathbf{u} \cdot \nabla)\mathbf{u}. \quad (2)$$

For incompressible flow ($\nabla \cdot \mathbf{u} = 0$), both forms are equivalent.

II. 2. PHYSICAL INTERPRETATION

The advection term represents the transport of momentum by the flow:

Momentum enters and leaves a control volume through its faces.

The net flux determines the rate of change of velocity.

III. 3. TENSOR FORMULATION

The outer product of velocity is:

$$\mathbf{u}\mathbf{u} = \begin{bmatrix} u^2 & uv & uv & v^2 \end{bmatrix} \quad (3)$$

Taking divergence:

u-momentum

$$-\left(\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y}\right) \quad (4)$$

v-momentum

$$-\left(\frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y}\right) \quad (5)$$

IV. 4. DISCRETIZATION STRATEGY

A. Step 1: Face Interpolation

$$u_{i-1/2,j} \approx \frac{u_{i-1,j} + u_{i,j}}{2} \quad (6)$$

B. Step 2: Flux Evaluation

$$(u^2) * i - 1/2, j \approx \left(\frac{u * i - 1, j + u_{i,j}}{2}\right)^2 \quad (7)$$

$$(uv) * i, j + 1/2 \approx \left(\frac{u * i, j + u_{i,j+1}}{2}\right) \left(\frac{v_{i-1,j+1} + v_{i,j+1}}{2}\right) \quad (8)$$

C. Step 3: Divergence of Fluxes

$$\frac{\partial u^2}{\partial x} \approx \frac{(u^2) * i + 1/2, j - (u^2) * i - 1/2, j}{\Delta x} \quad (9)$$

$$\frac{\partial uv}{\partial y} \approx \frac{(uv) * i, j + 1/2 - (uv) * i, j - 1/2}{\Delta y} \quad (10)$$

V. 5. DISCRETE ADVECTION OPERATOR

$$\mathcal{N}_u = -\left(\frac{\partial u^2}{\partial x} + \frac{\partial uv}{\partial y}\right) \quad (11)$$

$$\mathcal{N}_v = -\left(\frac{\partial uv}{\partial x} + \frac{\partial v^2}{\partial y}\right) \quad (12)$$

VI. 6. BOUNDARY CONDITION TREATMENT

Boundary conditions are directly embedded into flux computations.

A. Dirichlet Boundaries

At a boundary face:

$$u_{face} = \frac{u_{BC} + u_{interior}}{2} \quad (13)$$

B. Example (Bottom Boundary)

$$(uv) * i, 1/2 = u * BC(i) \cdot \frac{v_{BC}(i-1) + v_{BC}(i)}{2} \quad (14)$$

C. Key Idea

Missing stencil values are replaced with boundary values, eliminating the need for ghost nodes.

VII. 7. NUMERICAL PROPERTIES

A. Advantages

- Second-order accurate

- Conservative formulation
- Physically interpretable

B. Limitations

- Non-dissipative
- Can lead to oscillations at high Reynolds number

VIII. 8. ALGORITHM SUMMARY

1. Interpolate velocities to faces
2. Compute fluxes (u^2 , uv , v^2)
3. Compute differences in x and y
4. Assemble divergence
5. Apply boundary values

IX. 9. KEY TAKEAWAY

Advection is computed as the divergence of momentum fluxes across cell faces.